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textNet: Directed, Multiplex, Multimodal Event Network Extraction from Textual Data

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Introduction

The *textNet* package allows a user to input one or more PDF documents and create directed, multiplex, and multimodal network graphs. This enables analysis of the relationships between verb attributes and tenses, entity types, structural motifs, and other network characteristics. Entities mentioned within the input text become nodes, and the verbs connecting them in the sentences of the text become directed edges. Zufall and Scott (2024) demonstrate the use of *textNet* to identify which actors are involved in start-up versus ongoing management tasks, characterize patterns of information and funding flows, and compare the distribution of management tasks in networks from regions known to have contrasting characteristics.

textNet has applications in social science research, including governance network scholarship, as demonstrated by Zufall and Scott (2024) and by ongoing work on water resources governance and environmental impact assessments at the UC Davis Center for Environmental Policy and Behavior. *textNet* works on arbitrarily long documents, making it well suited for analyzing legal documents, court proceedings, plans and policy documents, media publications, and other long-form textual data containing events and entity relationships.

Statement of Need

Network measurement in social science typically relies on data collected through surveys and interviews. Document-based measurement can be automated and scaled, providing opportunities for large-N or longitudinal research that are infeasible through traditional methods. A number of tools exist to generate networks based on co-occurrence of words within documents (such as the Nocodefunctions app (Levallois et al., 2012), the "textnets" package (Bail, 2024), InfraNodus (Paranyushkin, 2018), and many more). However, existing network extraction methods that use co-occurrence leave a vast amount of data on the table, namely, the rich edge attribute data and directionality of each verb phrase defining the particular relationship between two entities, and the respective roles of the entity nodes involved in that verb phrase. There is, to our knowledge, no existing open-source tool that generates network data based on the syntactic relationships between entities within a sentence.

We present an R package, *textNet*, designed to enable directed, multiplex, and multimodal network extraction from text of documents through syntactic dependency parsing. The *textNet* package facilitates automated and replicable analysis and comparison of many documents, based on their respective network characteristics. Its flexibility allows for any desired entity categories, such as organizations, geopolitical entities, dates, or custom-defined categories, to be preserved.

Directed Graph Production

As a syntax-based network extractor, *textNet* identifies source and target nodes. This produces directed graphs that contain information about network flow. Methods based on identifying co-

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occurring nodes in a document, by contrast, produce undirected graphs. Co-occurrence graphs also tend to generate saturated subgraphs, since every co-occurring collection of entities has every possible edge drawn amongst them. By contrast, *textNet* draws connections specifically between pairs of entities that are mediated by an event relationship, rather than between every entity in the document or even in the sentence.

Multiplex Graph Output

Syntax-based measurement encodes edges based on subject-verb-object relationships. *textNet* stores verb information as edge attributes, which allows the user to preserve arbitrarily complex topological layers (of different types of relationships) or customize groupings of edge types to simplify representation.

Multimodal Graph Output

Multimodal networks, or networks where there are multiple categories of nodes, have common use cases such as social-ecological network analysis of configurations of actors and environmental features. Existing packages such as the *manynet* package (Hollway, 2024) provide analytical functions for multimodal network statistics. *textNet* provides a structure for tagging and organizing node labeling schemes that can then be fed into packages for multi-node network statistical analysis. Node labels can be automated (e.g., the default entity type tags for an NLP engine such as *spaCy* (Honnibal et al., 2021)), customized using a dictionary, or based on a hybrid scheme of default and custom labels. Any node type is possible (e.g., species, places, people, concepts, etc.) so this can be adapted to domain-specific research applications by applying dictionaries, as was used to define the "WATER" entity type in Figure 2 (see package vignette for details), or using a custom NER model.

Overview and Main Functions

The package architecture relies on four sets of functions around core tasks:

- [OPTIONAL] Preprocessing: `pdf_clean()`, a wrapper for the `pdftools::pdf_text()` function which includes a custom header/footer text removal feature; and `parse_text()`, which is a wrapper for the *spacyr* package and uses the *spaCy* natural language processing engine (Honnibal et al., 2021) to parse text and perform part of speech tagging, dependency parsing, and named entity recognition (NER). Alternatively, the user can skip this step and load parsed text directly into the package. Externally produced data must be converted to the format requirements outlined in the package manual.
- Network extraction: `textnet_extract()`, which generates a graph database from parsed text based upon tags and dependency relations. The object returned from `textnet_extract()` consists of a nodelist, an edgelist with a rich set of edge attributes, a verblist, and an appositivelist (containing potential coreferences such as acronyms and their full forms for disambiguation).
- Disambiguation: tools for cleaning, recoding, and aggregating node and edge attributes, such as the `find_acronyms()` function, which can be paired with the `disambiguation()` function to identify acronyms in the text and replace them with the full entity name.
- Exploration: the `export_to_network()` function for exporting the graph database to igraph and network objects, `top_features()` for viewing node and edge attributes, and `combine_networks()` for aggregating multiple document-based graphs based on common nodes.

The figure below summarizes the functionality of *textNet* and the flow of function outputs. Optional data cleaning features are shown with dotted arrows.

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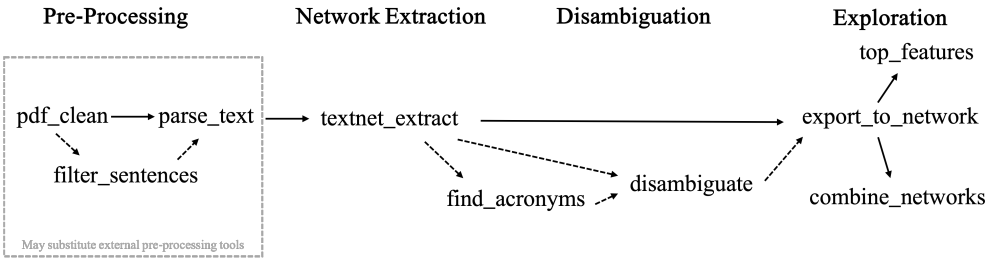


Figure 1: Workflow of *textNet* Functions

Installation

The stable version of this package can be installed from GitHub, using the *pak* package (Csárdi et al., 2026):

```
pak::pak("ucd-cepb/textNet")
```

The *textNet* package suggests several convenience wrappers of packages such as *spacyr* (Benoit et al., 2023), *pdftools* (Ooms, 2024), *igraph* (Csárdi et al., 2024), and *network* (Butts et al., 2023). To use the full functionality of *textNet*, such as preprocessing tools and post-processing analysis tools, we recommend installing these packages, which for *spacyr* requires integration with Python. However, the user may wish to preprocess and parse data using their own NLP engine, and skip directly to the `textnet_extract()` function, which does not depend on *spacyr* or Python integration.

Downstream Analysis

textNet is compatible with standard network analysis tools in R. Functionality provided by *ggraph* (Pedersen & RStudio, 2024), *sna* (Butts, 2024), *igraph* (Csárdi et al., 2024), *network* (Butts et al., 2023), and other network visualization and analysis packages can be used to further explore the extracted networks.

The *ggraph* package has been used to create the network visualization seen here, using a weighted version of an igraph constructed using the "example_plan_v2_parsed" sample data in *textNet*.

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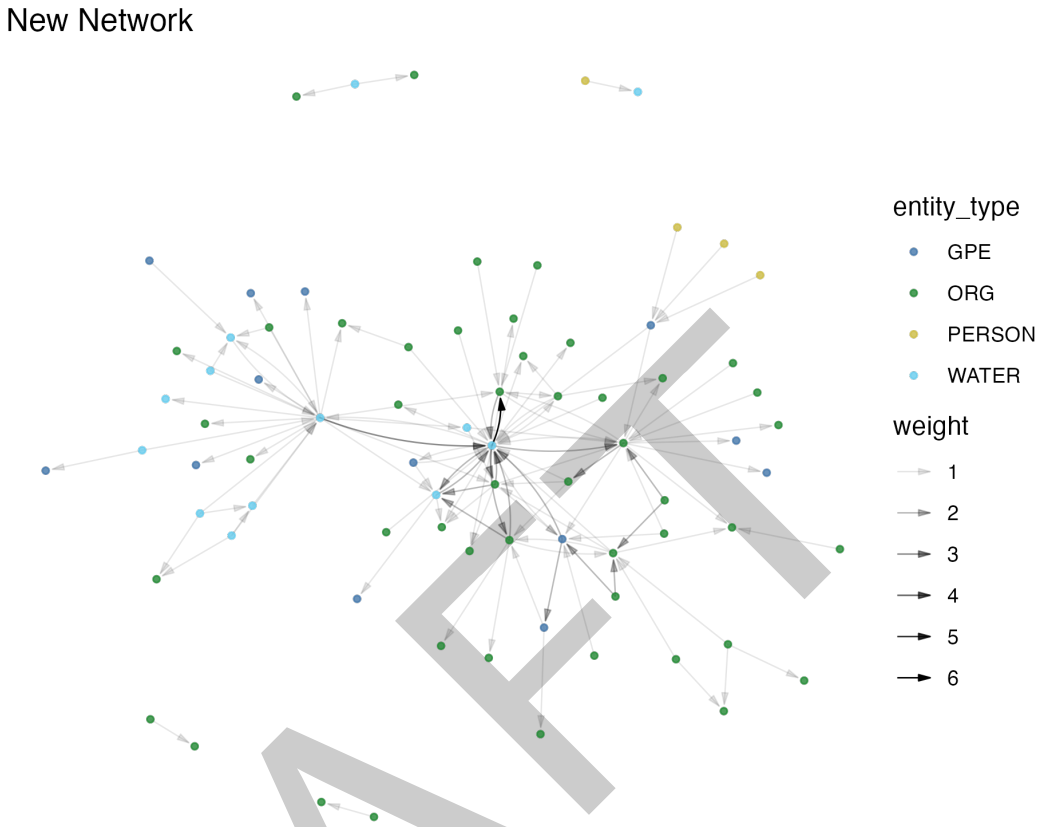


Figure 2: Representation of the Event Network of the New Plan

The network-level attributes output from `export_to_network()` can also be analyzed against exogenous metadata that has been collected separately by the researcher regarding the different documents and their real-world context. The extracted networks can also be analyzed through a variety of network analysis tools, such as an Exponential Random Graph Model or a Temporal Exponential Random Graph Model.

Vignette

More information about the entity network extraction algorithm and an example start-to-finish data processing and analysis workflow can be found in the vignette for this package. The vignette uses sample data that travels with the *textNet* package.

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